

Chicago Booth Accounting RP Data Task
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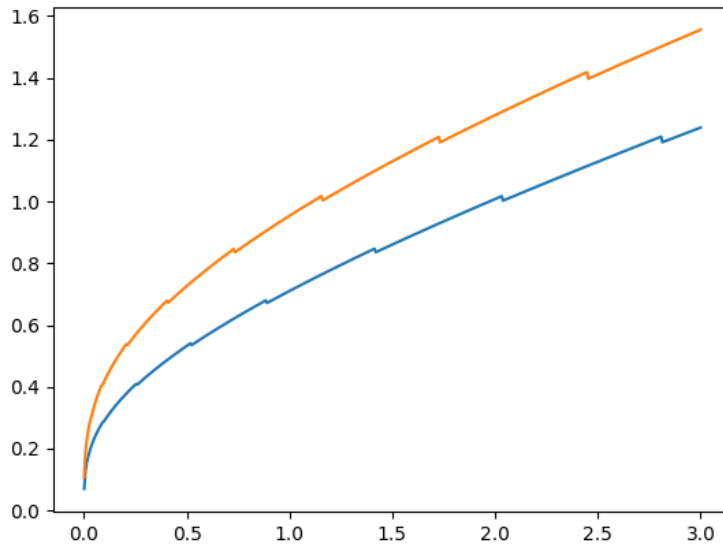
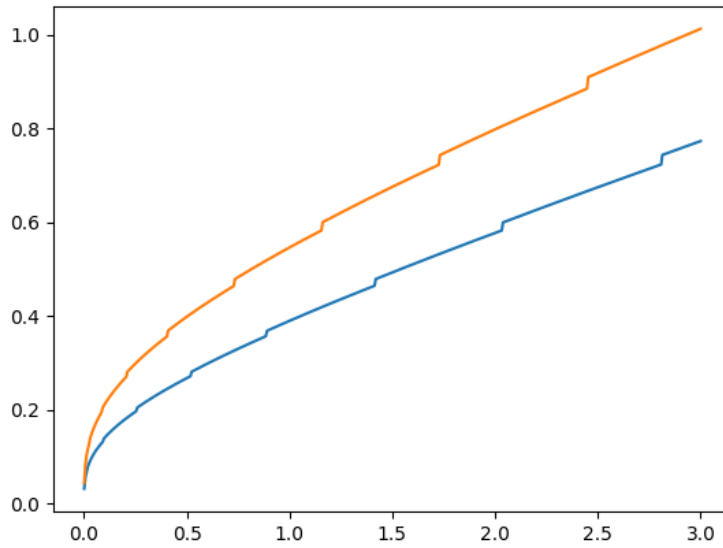
Solution (Stochastic Ramsey Problem).

1. Our Bellman equation is

$$\begin{aligned} V(K, z) &= \max_{K'} \{u(C) + \beta \mathbb{E}(V(K', z'))\} \\ &= \max_{K'} \left\{ u(F(K, z) - K') + \beta \left(\frac{1}{2} V(K', 1) + \frac{1}{2} V(K', 2) \right) \right\} \end{aligned}$$

where $u(x) = \log(x)$, $\beta = 0.9$, $F(K, z) = C + K'$ and $F(K, z) = A(z)K^{0.3} + 0.3K$.

2. See 'discretize.py' in folder for code. From bottom to top: Investment and consumption policy graphs on $[0.001, 3]$. Orange is when $z = 2$, blue is when $z = 1$.



3. The FOC is:

$$\begin{aligned} 0 &= \frac{d}{dK'} \log(F(K, z) - K') + \beta \left(\frac{1}{2} V(K', 1) + \frac{1}{2} V(K', 2) \right) \\ &= -\frac{1}{C} + \beta \mathbb{E} \left(\frac{d}{dK'} V(K', z') \right) \end{aligned}$$

The envelope condition is:

$$\begin{aligned} \frac{d}{dK} V(K, z) &= \frac{d}{dK} \log(F(K, z) - K') + \beta \left(\frac{1}{2} V(K', 1) + \frac{1}{2} V(K', 2) \right) \\ &= \frac{1}{C} \cdot \frac{d}{dK} F(K, z) \end{aligned}$$

And the Euler equation from the envelope condition on K one period ahead (i.e. $K' = K_{t+1}$):

$$\begin{aligned} \frac{1}{C} &= \beta \mathbb{E} \left(\frac{d}{dK'} V(K', z') \right) \rightsquigarrow \\ \frac{1}{C_t} &= \beta \mathbb{E} \left(\frac{1}{C_{t+1}} F(K_{t+1}, z_{t+1}) \right) \end{aligned}$$

Solution (Auditors and misstatements).

1. See 'Auditors_and_Restatements.ipynb' in folder for code.

In the 2008-2009 sample, Big 4 clients have a statistically significant (Chi-Squared, $p = .0102$) difference in restatement rate that is lower compared to non-Big 4 clients. However, this is correlational and there is no evidence in the dataset provided that the sample populations are not statistically different, i.e. that the result might reflect differences in the clientele. We can easily imagine, for instance, that Big 4 clients are some of the worlds largest, most complex, most visible and most well-off companies while non-Big 4 clients might, on average, be less so in all of these attributes. So even if it was the case that the auditors were the same quality, restatement rates among the latter population might be higher just because the companies are more error-prone. At the same time, though, Big 4 members are also more visible and thus more incentivized to not make mistakes. Regardless, there is no definite conclusion that can be drawn without more information. Furthermore, 2008-2009 is maybe an unusual period on account of the 2007 financial crisis, further confounding the analysis.

2. I'd expect that in three years the restatement rate would be higher, but the policy would bring restatement rates down in the long run. While in theory auditors could respond by practicing overcompliance, this seems unlikely. Over-compliance introduces more costs for the auditor which are offloaded to and costly to clients which also makes them more vulnerable to competition that are not practicing over-compliance. Auditors will try to guess at the new standard which leads to more errors because the incoming changes are unlikely to be perfectly understood and it takes time to build systems and train workers that do. So I expect that as practice improves because all auditors are more heavily incentivized to do so that the restatement rate will go down, but before and in the short run after the implementation, auditors are likely to miss the mark a bit.

3. See 'Auditors_and_Restatements.ipynb' in folder for code.

Table on next page.

SectorId	SectorName	RestatementType1	CompanyCount	CompanyID	CompanyName	FirstRestateDate	MinDays	MedianDays	MaxDays
1	Basic Materials	ER	8	1096841	Case Financial Inc.	2008-01-14 00:00:00	8.00	30.00	255.00
1	Basic Materials	OTH	27	1288750	Timberline Resources Corp.	2008-01-14 00:00:00	0.00	9.00	700.00
1	Basic Materials	RR	2	1014669	Sterling Chemicals, Inc.	2008-04-11 00:00:00	209.00	209.00	209.00
2	Capital Goods	ER	6	357108	Irvine Sensors Corp.	2008-01-15 00:00:00	0.00	11.00	136.00
2	Capital Goods	OTH	20	80035	Preformed Line Products Co.	2008-01-04 00:00:00	1.00	13.00	675.00
2	Capital Goods	RR	2	65201	Met-Pro Corporation	2008-02-12 00:00:00	220.00	220.00	220.00
3	Consumer Cyclical	ER	4	1160858	True Religion Apparel Inc.	2008-05-01 00:00:00	60.00	106.00	547.00
3	Consumer Cyclical	OTH	21	923168	IWI Holding Ltd.	2008-01-09 00:00:00	0.00	12.50	582.00
3	Consumer Cyclical	RR	1	1046692	LJ International Inc.	2008-06-19 00:00:00	NaN	NaN	NaN
4	Consumer Non-Cyclical	ER	2	910406	Hain Celestial Group Inc.	2008-01-31 00:00:00	242.00	242.00	242.00
4	Consumer Non-Cyclical	OTH	11	707674	FACT Corp.	2008-02-13 00:00:00	14.00	24.00	407.00
4	Consumer Non-Cyclical	RR	4	835011	MGP Ingredients, Inc.	2008-05-12 00:00:00	35.00	113.00	323.00
5	Energy	ER	4	106455	Westmoreland Coal Co.	2008-03-17 00:00:00	29.00	55.00	620.00
5	Energy	OTH	28	924805	BMB Munai Inc.	2008-01-10 00:00:00	0.00	9.00	708.00
5	Energy	RR	5	789606	Aztec Oil & Gas Inc.	2008-01-03 00:00:00	26.00	95.50	582.00
6	Financial	ER	7	354647	CVB Financial Corp.	2008-02-29 00:00:00	0.00	15.50	367.00
6	Financial	OTH	27	17313	Capital Southwest Corp.	2008-01-09 00:00:00	0.00	3.00	576.00
6	Financial	RR	1	100716	UNICO AMERICAN CORP	2008-08-13 00:00:00	NaN	NaN	NaN
7	Healthcare	ER	9	57055	KV Pharmaceutical Co.	2008-03-25 00:00:00	3.00	21.00	601.00
7	Healthcare	OTH	40	1057698	MDS Inc.	2008-01-29 00:00:00	0.00	7.00	659.00
7	Healthcare	RR	4	877357	Sepracor Inc.	2008-02-29 00:00:00	17.00	30.00	175.00
8	Services	ER	14	1123870	CIBT Education Group Inc.	2008-01-02 00:00:00	2.00	20.00	677.00
8	Services	OTH	79	37008	Winthrop Realty Trust	2008-01-07 00:00:00	0.00	3.00	668.00
8	Services	RR	8	891103	IAC Interactive	2008-02-29 00:00:00	12.00	20.00	398.00
9	Technology	ER	15	855906	Silicon Storage Technology Inc.	2008-01-16 00:00:00	2.00	15.00	258.00
9	Technology	OTH	46	23082	Computer Sciences Corp.	2008-01-11 00:00:00	0.00	6.00	629.00
9	Technology	RR	10	816761	Teradata Corporation	2008-03-03 00:00:00	1.00	42.00	605.00